Digital machining database for machine learning

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Abstract
The purpose of the open-source Digital Machining Database is to support machine learning algorithm development for machining process modeling. The database is made available to the research community and provides organized data for algorithm training, validation, and benchmarking. The data are generated using deterministic, physics-based numerical (time domain) and analytical (frequency domain) simulations for many milling conditions. For each dataset corresponding to a selected set of milling conditions, hundreds of simulation results are provided for individual spindle speed-axial depth of cut combinations. The time domain signals are: 1) tool displacement, velocity, and acceleration; 2) workpiece displacement, velocity, and acceleration; and 3) cutting force. The spindle speed-dependent stability boundary is also included for each dataset.

Keywords
Machining, machine learning, simulation, modeling

1. Introduction

Machine learning (ML) represents a subset of artificial intelligence (AI) where computers learn from data without explicit programming. Algorithms and statistical models are applied to analyze data patterns and identify relationships based on these patterns. Machine learning is in its early stages of application within manufacturing environments, but many opportunities are available. These include process improvement, quality control, energy efficiency, cost savings, predictive maintenance, supply chain optimization, customization and personalization, and safety, to name a few [1-2].

Many instances of ML for machining have been presented in the literature. Tnani et al. provided a dataset for process monitoring of milling machines based on acceleration data [3]. Manikanta et al. reviewed applications of AI and ML in machining operations, including models for cutting forces, tool wear, energy consumption, and machining parameter optimization [4]. Kaur et al. summarized AI, ML, and deep learning implementations for machining processes and identified research gaps such as data quality, system adaptability, and scalability [5]. Liang et al. detailed applications of ML to precision machining with a focus on model architectures and methodologies for industrial implementation [6]. Many other publications are available and the number of submissions continues to increase.

To support the machining community's growing interest in developing physics-informed/physics-guided ML models by combining physics-based, deterministic machining models with data-driven models, this paper describes an open-source Digital Machining Database [7] hosted by Kaggle, an online platform that serves as a hub for the data science and ML communities. Data challenges for manufacturing environments include: 1) open access to large volumes of process data; 2) the cost of data generation and storage; 3)

availability of equipment and sensors to generate data; 4) the often-proprietary nature of production data; and 5) the quality of the available data. To address these challenges and enable new model development, validation, and benchmarking, the Digital Machining Database currently includes 50 milling datasets, each with its own set of tool and workpiece dynamics, cutting force model coefficients, cut geometry, and tool geometry. As community interest grows, the database will be expanded to include additional datasets with new sets of milling conditions.

In each dataset, time domain tool and workpiece vibration (displacement, velocity, and acceleration) and cutting force signals are provided for hundreds of spindle speed-axial depth of cut combinations. The spindle speed-dependent stability boundary is also included for reference. The database enables algorithm and model development for the prediction of milling cutting forces, tool and workpiece vibrations, stability for chatter avoidance, and surface location error [8], for example. Details about the database are provided in Sections 2 and 3.

2. Simulation inputs

As noted in Section 1, time domain simulation is applied to predict milling signals, which are made available in individual datasets for training ML models. The numerical solution of the time-delay, second-order set of differential equations of motion that describe the milling process is detailed by Schmitz and Smith [9]. The spindle speed-dependent stability boundary for each dataset is calculated using the zero-order frequency domain solution presented by Altintas and Budak [10]. The simulation inputs are the tool tip frequency response functions (FRFs), workpiece FRFs, mechanistic cutting force model coefficients, tool description, and cut description. These inputs are detailed in Sections 2.1 to 2.4 and are provided for each dataset in a spreadsheet included with the database.

2.1 FRFs

The tool tip (direct) FRF modal parameters for the *x*, or feed, and *y* directions are defined, where *z* is aligned with the cutting tool's axis. The {natural frequency, modal stiffness, and viscous damping ratio} triplet is specified for each vibration mode in the *x* and *y* directions. The same format is used for the workpiece FRFs.

2.2 Cutting force model coefficients

The tangential (*t*) and normal (*n*) direction cutting force, *F*, component models are given by Eqs. 1 and 2, where each includes both a cutting/shearing (*c*) and edge/rubbing (*e*) term with two associated coefficients, *k*; *b* is the axial depth of cut; and *h* is the instantaneous chip thickness, which depends on the tooth angle, ϕ, and feed per tooth [9]. See Fig. 1 for the force geometry. The four cutting force model coefficients are provided for each dataset.

$$F_t = k_{tc} b h + k_{te} b \tag{1}$$

$$F_n = k_{nc} b h + k_{ne} b \tag{2}$$

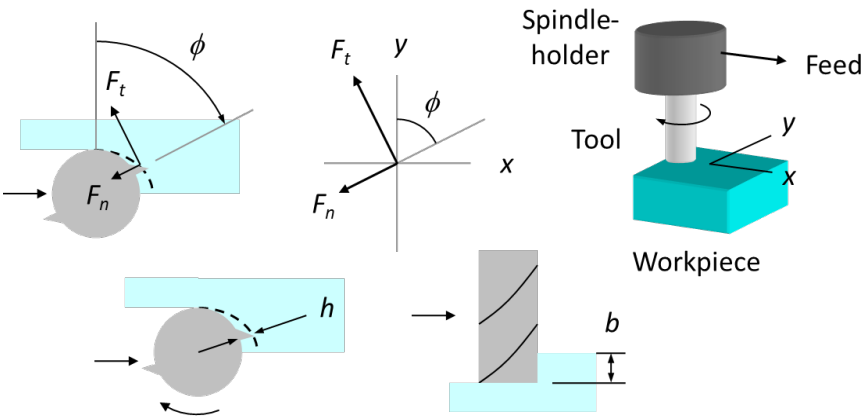


Figure 1: Simulation force modeling.

Process damping is also included in the time domain simulation using the force model provided by Eq. 3, where C_n is the normal direction process damping coefficient, v is the cutting speed, and $\dot{n}$ is the relative velocity between the tool and workpiece in the normal direction. The process damping force from Eq. 3 is summed with the normal direction cutting force component from Eq. 2. The process damping coefficient is also provided for each dataset.

$$F_d = -C_n \frac{b}{v} \dot{n} \tag{3}$$

2.3 Tool description

The tool description includes the diameter, number of teeth, helix angle vector with one entry for each tooth, tooth angle vector with one entry for each tooth, runout vector with one entry for each tooth, and the endmill type, including square, ball nose, or bull nose. This format incorporates variable helix, variable tooth spacing, and tooth-to-tooth runout effects in the time domain simulation results.

2.4 Cut description

The cut description includes the cut direction (up/conventional or down/climb milling), radial depth of cut, axial depth of cut, commanded feed per tooth, and spindle speed.

3. Simulation outputs

As noted in Section 1, time domain simulation is applied to calculate milling signals including: 1) tool displacement, velocity, and acceleration; 2) workpiece displacement, velocity, and acceleration; and 3) cutting force. The time domain outputs are saved in two formats: .mat (MATLAB) and .h5 (hierarchical data format version 5 for storage and organization of large datasets, often selected for Python applications). The naming convention for the individual data files is timeA_B.mat and timeA_B.h5, where A is the dataset number and B is the grid point number. The grid point numbers are ordered from the lowest spindle speed (lowest to highest axial depth at that speed) to the highest spindle speed (lowest to highest axial depth at that speed) in the spindle speed-axial depth domain. Each time domain data file includes:

- spindle speed
- axial depth of cut
- 15 time domain vectors
    - discretized time

- tool displacement in the *x* direction
- tool displacement in the *y* direction
- workpiece displacement in the *x* direction
- workpiece displacement in the *y* direction
- tool velocity in the *x* direction
- tool velocity in the *y* direction
- workpiece velocity in the *x* direction
- workpiece velocity in the *y* direction
- tool acceleration in the *x* direction
- tool acceleration in the *y* direction
- workpiece acceleration in the *x* direction
- workpiece acceleration in the *y* direction
- tool cutting force in the *x* direction
- tool cutting force in the *y* direction.

Figure 2 displays the spindle speed-axial depth coordinates for the Dataset 1 time domain data files as circles and demonstrates the data file numbering scheme. Each data file may represent stable or unstable (chatter) milling conditions, depending on the spindle speed-axial depth combination. The time domain outputs are not labeled as stable or unstable. This activity is left to the ML modeler.

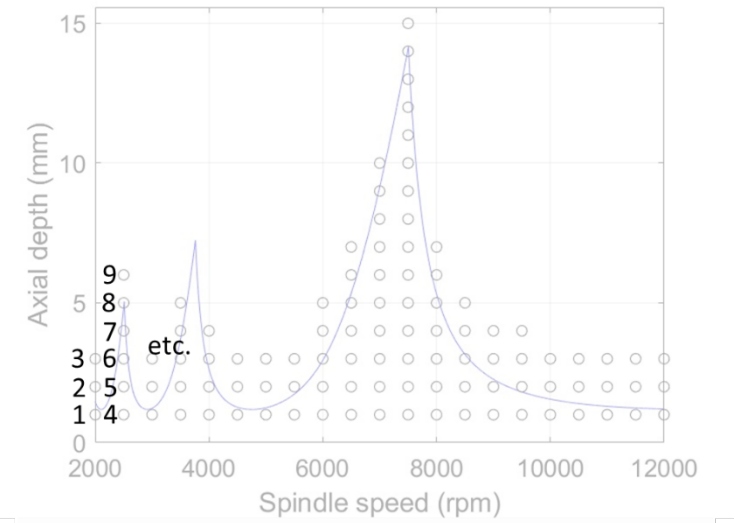


Figure 2: Spindle speed-axial depth combination coordinates (circles) for the individual time domain data files included in Dataset 1. The numbering scheme is demonstrated. The stability boundary is also included (blue curve).

Additionally, Fig. 2 shows the Dataset 1 stability boundary as the blue curve [9]. The file naming convention is stability_boundaryA.mat and stability_boundaryA.h5, where A is the dataset number. Each stability boundary data file includes two vectors:

- spindle speed
- stability boundary (i.e., the limiting axial depth at each spindle speed to avoid chatter).

Figure 3 displays two of the available time domain signals for a single spindle speed-axial depth combination from Dataset 29. As seen, Dataset 29 provides 1071 files for distinct combinations of spindle speed-axial depth and includes a non-zero process damping coefficient in the cutting force model (see Section 2.2). The right image shows the 1071 spindle speed-axial depth combinations as circles and the stability boundary as the blue curve (the stability boundary does not include process damping effects). The left image shows the time domain *x* direction tool tip displacement, x_t, and *x* direction cutting force, F_x, for the {1000 rpm, 3 mm} combination. Subsets of the complete time signals are shown for illustration.

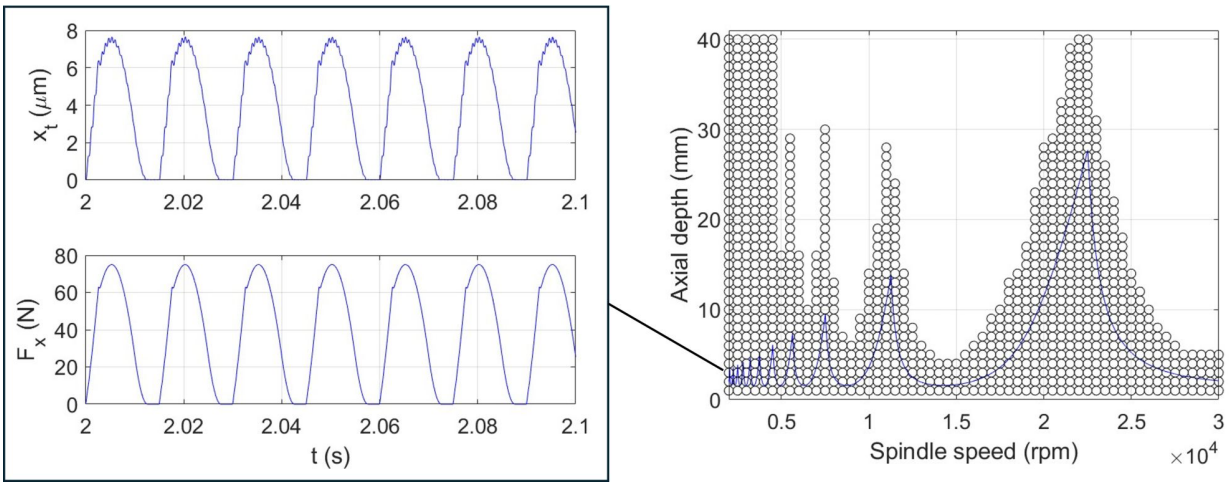


Figure 3: Example for Dataset 29. (Left) The time domain *x* direction tool tip displacement (top) and cutting force (bottom) for the {1000 rpm, 3 mm} data file. (Right) The 1071 spindle speed-axial depth combinations are superimposed as circles on the stability boundary (blue curve).

MATLAB [10] and Python [12-13] code examples are provided within the database for loading and plotting the individual data files from each dataset. There are currently 50 datasets available in the database. As noted, each dataset is based on a set of milling conditions and the data files within each dataset represent either time domain simulation results at a selected spindle speed-axial depth combination or the spindle speed-dependent stability boundary. A spreadsheet is included in the database that provides the milling conditions (inputs) for each of the 50 datasets following the definitions from Section 2.

Conclusions

This paper described the open-source Digital Machining Database [7]. The purpose of the database is to support machine learning algorithm development for machining process modeling by providing a comprehensive, no-cost, non-proprietary data source. It is made available for download via Kaggle, an online platform that serves as a hub for the data science and machine learning communities, and provides organized data for algorithm training and benchmarking. The data are generated using numerical (time domain) and analytical (frequency domain) simulations for various milling conditions. There are currently 50 datasets in the database with hundreds of time domain data files within each dataset for various spindle speed-axial depth combinations, as well as the stability boundary. New datasets will be added based on the modeling community's feedback and requests.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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